R Lab: Week 9

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```
library(dplyr, warn.conflicts = F)
library(stringr)
library(ggplot2)
library(readr)
library(qdapDictionaries)
library(purrr, warn.conflicts = F)
```

Today we we will study the wage gap between men and women among Chicago's public employees and learn a bit about matrix notation in R.

The following code reads in data from Open Chicago, a data source on Chicago city information.

```
employees <- read_csv("~/Dropbox/mathCamp/Code/data/CurrentEmployees.csv", col_types = "cccc")
employees <- employees %>% mutate(salarynum = as.numeric(str_extract(`Employee Annual Salary`,
    "[0-9]+")), nolastname = str_extract(Name, "(?<=,).+"), firstname = str_extract(nolastname,
    "[^\\s]+"))
```

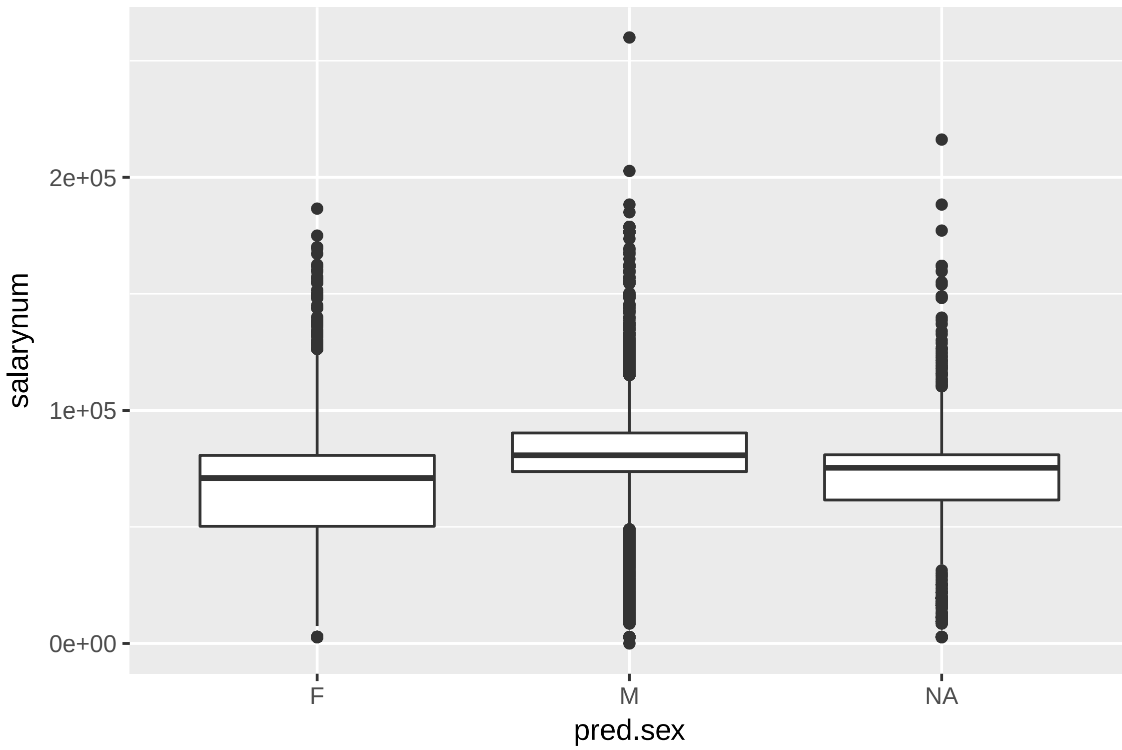
The qdapDictionaries library has a file called NAMES, which include columns including name and pred.sex, which is the predicted binary gender of each name. Here we join by exact matches.

```
UNAMES <- unique(NAMES[, c("name", "pred.sex", "gender2")])
names(UNAMES) <- c("firstname", "pred.sex", "gender2")
codedgender <- employees %>% left_join(UNAMES)
```

```
## Joining, by = "firstname"
```

From here we can begin to study the differences in salary across genders.

```
ggplot(codedgender, aes(pred.sex, salarynum)) + geom_boxplot()
```




```
justcoded <- codedgender %>% select(pred.sex, salarynum) %>% na.omit()

justcoded %>% group_by(pred.sex) %>% summarise(meansalary = mean(salarynum), sdsalary = sd(salarynum))
```

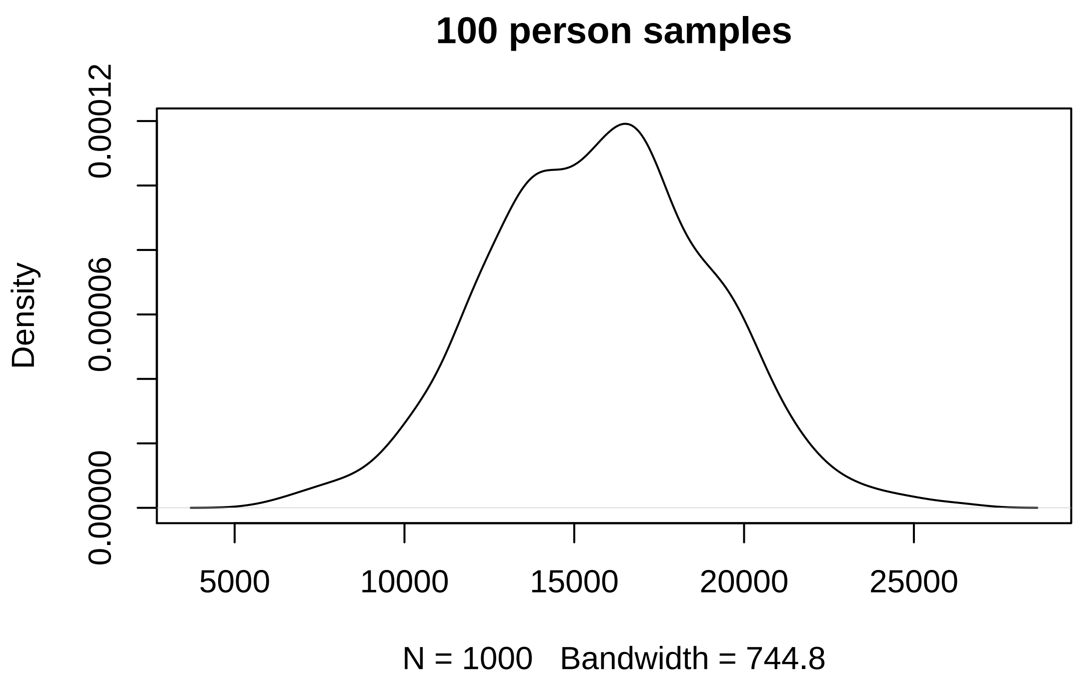
```
## # A tibble: 2 × 3
##   pred.sex meansalary sdsalary
##     <fctr>      <dbl>    <dbl>
## 1        F   64993.34 27804.26
## 2        M   80837.92 19129.96
```

```
meansamples <- c()
for (i in 1:1000) {

    mensalarys100 <- sample(justcoded$salarynum[justcoded$pred.sex == "M"], 100)
    womensalarys100 <- sample(justcoded$salarynum[justcoded$pred.sex == "F"], 100)

    meansamples[i] <- mean(mensalarys100) - mean(womensalarys100)
}

plot(density(meansamples), main = "100 person samples")
```

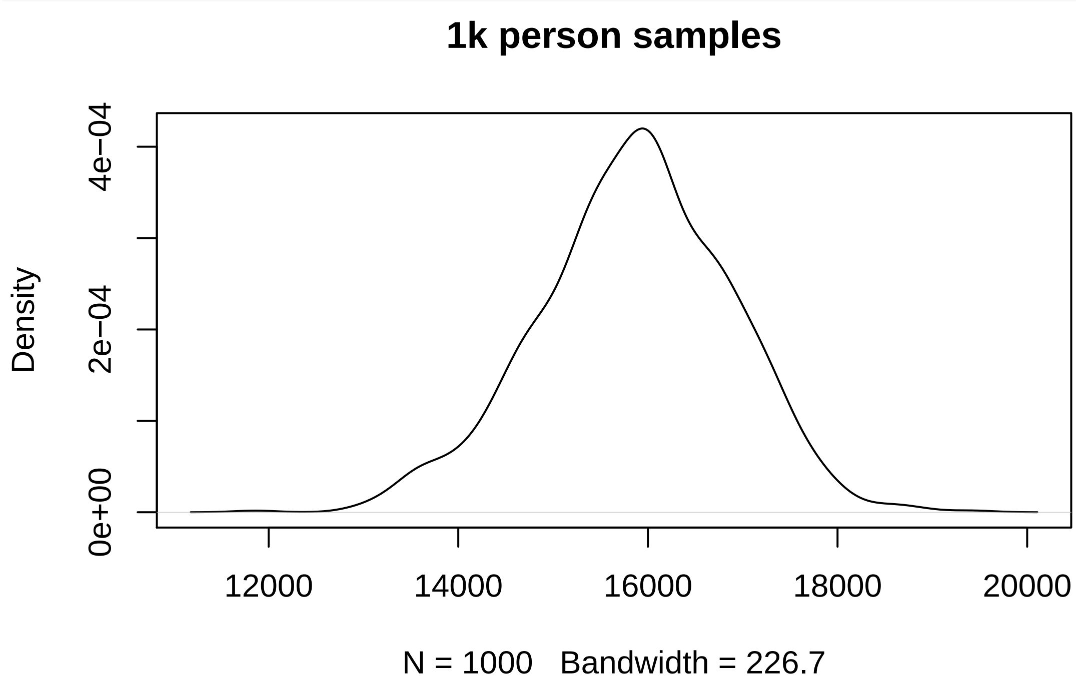



```
meansamples1k <- c()
for (i in 1:1000) {

    mensalarys1k <- sample(justcoded$salarynum[justcoded$pred.sex == "M"], 1000)
    womensalarys1k <- sample(justcoded$salarynum[justcoded$pred.sex == "F"], 1000)

    meansamples1k[i] <- mean(mensalarys1k) - mean(womensalarys1k)
}

plot(density(meansamples1k), main = "1k person samples")
```



We can do similar analysis Using (prehaps too much) dplyr

```
ttestout<-justcoded %>%

  summarise(

  gapestimate = diff(t.test(salarynum[pred.sex == "M"],
                 salarynum[pred.sex == "F"],
                 var.equal = T )$estimate),

  tstat = t.test(salarynum[pred.sex == "M"],
                 salarynum[pred.sex == "F"],
                 var.equal = T )$statistic,

  df = t.test(salarynum[pred.sex == "M"],
              salarynum[pred.sex == "F"],
              var.equal = T )$parameter,

  pvalue=1-pt(tstat, df))

ttestout
```

```
## # A tibble: 1 × 4
##   gapestimate    tstat    df pvalue
##         <dbl>    <dbl> <dbl>  <dbl>
## 1   -15844.58 55.85816 29098      0
```

```
summary(lm(salarynum ~ pred.sex, data = justcoded))
```

```
##
## Call:
## lm(formula = salarynum ~ pred.sex, data = justcoded)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -80838  -9090   -114  12354 179166
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  64993.3      238.4  272.57   <2e-16 ***
## pred.sexM    15844.6      283.7   55.86   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 22030 on 29098 degrees of freedom
## Multiple R-squared:  0.09684,    Adjusted R-squared:  0.09681
## F-statistic:  3120 on 1 and 29098 DF,  p-value: < 2.2e-16
```

It seems that women earn 15,800 dollars less, on average, than men.

Now we might be worried that we have unclassified men and women. Suppose we had a simple model for names, where $p(\theta_i|\theta_{i-1})$ is the probability of a letter i following letter $i-1$. We can then populate a matrix of letter to letter transitions. For example, is a "B" likely to follow a "Q"? Probably not. I add two spaces for the first and last letter respectively. We can then estimate these probabilities from the labeled names.

```
Female <- UNAMES %>% filter(pred.sex == "F")
lettersinnamesF <- str_split(Female$firstname, "")

lettersmat <- matrix(rep(0, 28 * 28), nrow = 28)

pupdate <- function(lnames) {
    letternums <- c(27, unlist(map(lnames, function(x) {
        which(LETTERS == x)
    })))
    letternumfollow <- c(letternums[-1], 28)
    lettersmat[letternums, letternumfollow] <- 1
    return(lettersmat)
}

mappingsF <- map(lettersinnamesF, pupdate)


finalmatF <- Reduce("+", mappingsF)

letterrowsumF <- rowSums(finalmatF)

finalnormF <- finalmatF/letterrowsumF
```

Here we have the full transition matrix for female names.

We can repeat the same process for male names.

```
Male <- UNAMES %>% filter(pred.sex == "M")
lettersinnamesM <- str_split(Male$firstname, "")

lettersmat <- matrix(rep(0, 28 * 28), nrow = 28)

pupdate <- function(lnames) {
    letternums <- c(27, unlist(map(lnames, function(x) {
        which(LETTERS == x)
    })))
    letternumfollow <- c(letternums[-1], 28)
    lettersmat[letternums, letternumfollow] <- 1
    return(lettersmat)
}

mappingsM <- map(lettersinnamesM, pupdate)

finalmatM <- Reduce("+", mappingsM)

letterrowsumM <- rowSums(finalmatM)
finalnormM <- finalmatM/letterrowsumM
```

Ok, so for the two genders, the probability of each letter following a B is:

```
finalnormM[2, ]
```

```
##  [1] 0.060723514 0.158914729 0.016795866 0.025839793 0.089147287
##  [6] 0.005167959 0.005167959 0.009043928 0.043927649 0.002583979
## [11] 0.009043928 0.040051680 0.010335917 0.054263566 0.062015504
## [16] 0.001291990 0.000000000 0.109819121 0.015503876 0.056847545
## [21] 0.031007752 0.000000000 0.005167959 0.000000000 0.028423773
## [26] 0.000000000 0.000000000 0.158914729
```

```
finalnormF[2, ]
```

```
##  [1] 0.0909614277 0.1560161197 0.0103626943 0.0213010938 0.1105354059
##  [6] 0.0011514105 0.0092112838 0.0213010938 0.0759930915 0.0000000000
## [11] 0.0120898100 0.0529648820 0.0161197467 0.0500863558 0.0230282096
## [16] 0.0017271157 0.0000000000 0.0817501439 0.0178468624 0.0408750720
## [21] 0.0115141048 0.0028785262 0.0005757052 0.0000000000 0.0310880829
## [26] 0.0046056419 0.0000000000 0.1560161197
```

Awsome huh? Notice that there is 0 probability of the character reserved for the beginning of a word following a B. Checking our intuition from earlier, the probablity that a "B" follows a "Q" is rightfully 0. Now we can see if this works. Suppose we test the accuracy.

```
namestest <- c(sample(Male$firstname, 500), sample(Female$firstname, 500))

namecut <- str_split(namestest, "")

pcheck <- function(lnames) {
    letternums <- c(27, unlist(map(lnames, function(x) {
        which(LETTERS == x)
    })))
    letternumfollow <- c(letternums[-1], 28)
    fprob <- prod(diag(finalnormF[letternums, letternumfollow]))
    mprob <- prod(diag(finalnormM[letternums, letternumfollow]))
    fprob <- fprob/(fprob + mprob)
    return(fprob)
}

probs <- unlist(map(namecut, pcheck))

table(c(rep("M", 500), rep("F", 500)), ifelse(probs > 0.5, "Flab", "Mlab"))
```

```
##
##     Flab Mlab
##   F  367  133
##   M  166  334
```

Seems that we don't always get it wrong! My own name (BOBBY) occurs as a male name with probability 0.9044971. ROBERT is male with probability 0.8309851.

The most male name in the dataset is MAXWELL, the least male name (of male names) is NATHANIAL.

```
Male$probF <- unlist(map(str_split(Male$firstname, ""), pcheck))
Male %>% arrange(probF) %>% head(3)
```

```
##   firstname pred.sex gender2       probF
## 1   MAXWELL        M       M 0.000000000
## 2   WOODROW        M       M 0.008434792
## 3  WILFREDO        M       M 0.019605808
```

```
Male %>% arrange(probF) %>% tail(3)
```

```
##      firstname pred.sex gender2     probF
## 1144    ISAIAH        M       M 0.9022252
## 1145    ISAIAS        M       M 0.9027460
## 1146 NATHANIAL        M       M 0.9102413
```

Among the most Female name is JACKQUELINE, the least Female Female name is WILLOW, followed by BUFFY (coincidence?).

```
Female$probF <- unlist(map(str_split(Female$firstname, ""), pcheck))
Female %>% arrange(probF) %>% tail(3)
```

```
##        firstname pred.sex gender2 probF
## 4014        ROXY        F       F     1
## 4015        XUAN        F       F     1
## 4016 JACKQUELINE        F       F     1
```

```
Female %>% arrange(probF) %>% head(3)
```

```
##   firstname pred.sex gender2      probF
## 1    WILLOW        F       F 0.03935616
## 2     BUFFY        F       F 0.05582483
## 3   MARJORY        F       F 0.07290241
```

We can use a more conservative coding to try to rule out making guesses.

```
namestest <- c(sample(Male$firstname, 500), sample(Female$firstname, 500))

namecut <- str_split(namestest, "")
probs <- unlist(map(namecut, pcheck))

table(c(rep("M", 500), rep("F", 500)), ifelse(probs > 0.6, "Flab", ifelse(probs <
    0.4, "Mlab", "Nolab")))
```

```
##
##      Flab Mlab Nolab
##   F   271   75   154
##   M    86  233   181
```

Great, this way we get substantially lower errors, at the cost of not coding about a third of the data. This indicates a tradeoff - we can either make more mistakes, or we can leave out data. Note, this is still not perfect!

Using this formulation, we can try to classify the uncoded genders as follows:

```
uncoded <- codedgender %>% filter(!pred.sex %in% c("F", "M"))
namecut <- str_split(uncoded$firstname, "")
uncoded$probsF <- unlist(map(namecut, pcheck))

uncoded %>% select(Name, firstname, probsF) %>% arrange(probsF) %>% head()
```

```
## # A tibble: 6 × 3
##                     Name firstname     probsF
##                    <chr>     <chr>      <dbl>
## 1 DEL MUNDO,  ROCKWELL R  ROCKWELL 0.00000000
## 2            HO,  KWOK L      KWOK 0.00000000
## 3      MILLAN,  BJORNN M    BJORNN 0.00000000
## 4 NIZAMUDDIN,  SHOWKET F   SHOWKET 0.00000000
## 5     LITTLE,  DURWARD J   DURWARD 0.02131448
## 6   RODRIGUEZ,  SIGFREDO  SIGFREDO 0.02402575
```

```
uncoded %>% select(Name, firstname, probsF) %>% arrange(desc(probsF)) %>% head()
```

```
## # A tibble: 6 × 3
##                       Name  firstname probsF
##                      <chr>      <chr>  <dbl>
## 1          BALATA,  TOMASZ     TOMASZ      1
## 2 BANACHOWSKI,  FRANCISZEK FRANCISZEK      1
## 3      CHLEBOWICZ,  LESZEK     LESZEK      1
## 4    CHOJNACKI,  MARIUSZ A    MARIUSZ      1
## 5      CUNNINGHAM,  RAQHEL     RAQHEL      1
## 6        CWIAKALA,  TOMASZ     TOMASZ      1
```

```
summary(uncoded$probsF)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's 
##  0.0000  0.3721  0.5569  0.5363  0.7015  1.0000       4
```

Finally, we can say, for those that are more than 60 percent likely to be male, call them male, for those that are more than 60 percent likely to be female, call them female.

```
namecut <- str_split(codedgender$firstname, "")
codedgender$probsF <- unlist(map(namecut, pcheck))
codedgender <- codedgender %>% mutate(newGender2 = ifelse(pred.sex %in% c("M", "F"),
    as.character(pred.sex), ifelse(probsF > 0.6, "F", ifelse(probsF < 0.4, "M", NA))))
table(codedgender$newGender2, codedgender$pred.sex, useNA = "always")
```

```
##
##            F     M  <NA>
##   F     8537     0  1311
##   M        0 20563   886
##   <NA>     0     0   863
```

```
summary(lm(salarynum ~ newGender2 == "F", data = codedgender))
```

```
##
## Call:
## lm(formula = salarynum ~ newGender2 == "F", data = codedgender)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -80582  -9494    142  12610 179422
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)            80581.6      151.9  530.50   <2e-16 ***
## newGender2 == "F"TRUE -15558.2      270.8  -57.46   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 22250 on 31295 degrees of freedom
##   (863 observations deleted due to missingness)
## Multiple R-squared:  0.09542,    Adjusted R-squared:  0.09539
## F-statistic:  3301 on 1 and 31295 DF,  p-value: < 2.2e-16
```

OK, so given all of this, can we simulate a male and female name? YES!

Lets find two names that start with the letter M.

```
simMname <- "M"

letterdraw <- "M"
for (i in 1:10) {
    if (letterdraw == ".") {
        break
    } else {
        letterdraw <- sample(c(LETTERS, "."), prob = finalnormM[which(LETTERS ==
            letterdraw), c(1:26, 28)], size = 1)
        simMname <- paste(simMname, letterdraw, sep = "")
    }
}
simMname
```

```
## [1] "MEDOU."
```

```
simFname <- "M"

letterdraw <- "M"
for (i in 1:10) {
    if (letterdraw == ".") {
        break
    } else {
        letterdraw <- sample(c(LETTERS, "."), prob = finalnormF[which(LETTERS ==
            letterdraw), c(1:26, 28)], size = 1)
        simFname <- paste(simFname, letterdraw, sep = "")
    }
}
simFname
```

```
## [1] "MASIL."
```

I would not name my kid MEDOU. or MASIL., but if you did, it would be a very masculine/feminine name (according to our simple model).